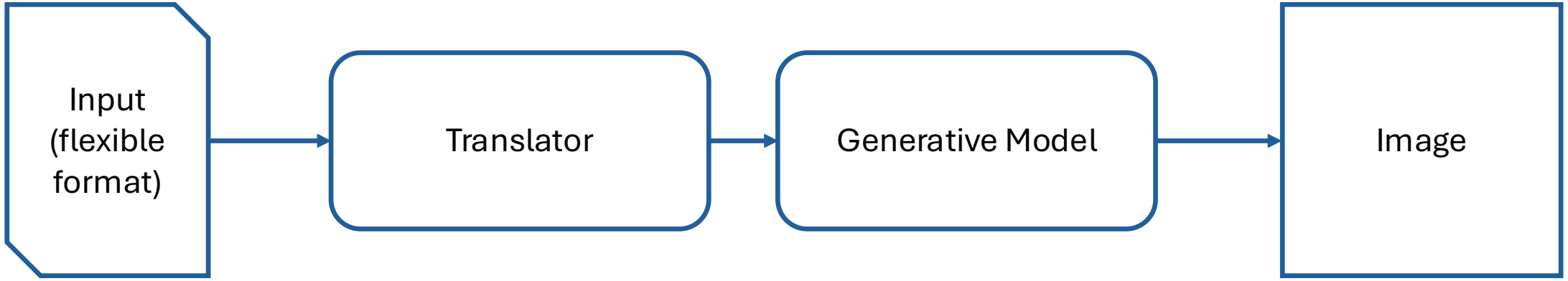
ECE 590
Neural-Network-Based Large Language Models

Tuesdays and Thursdays 1:25-2:40
Fitzpatrick Schiciano A 1464

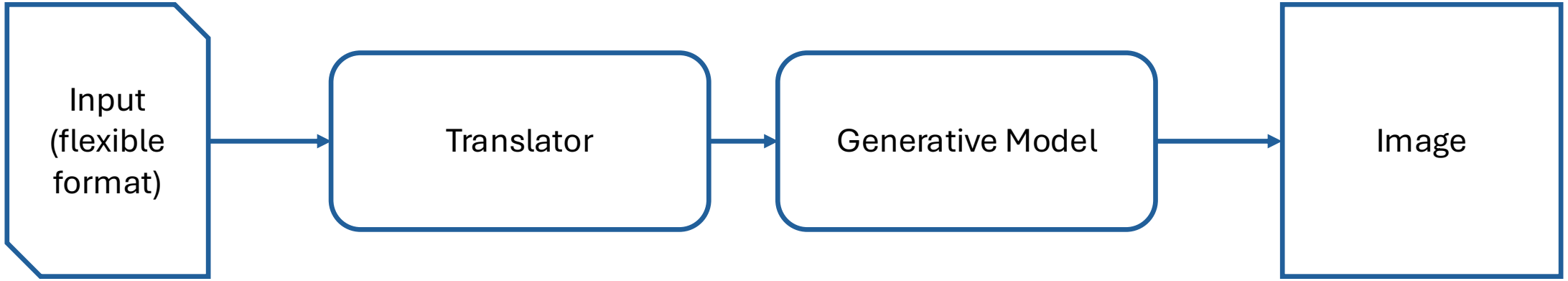
ECE 590
Neural-Network-Based Large Language Models

Lecture 23
04/09/2026

Desired Behavior when Generating Images

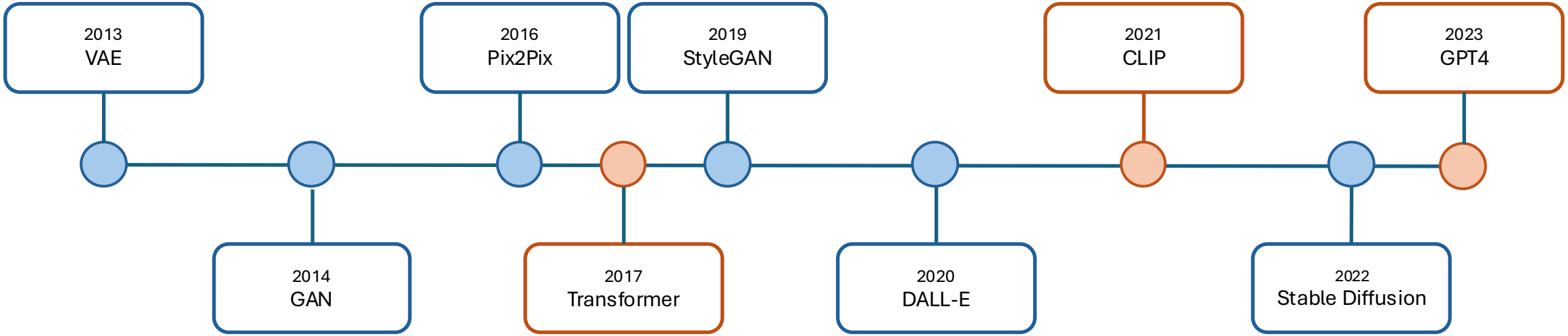


Desired Behavior



Generate an image of the duke machine learning school

Timeline of Image Generation Models



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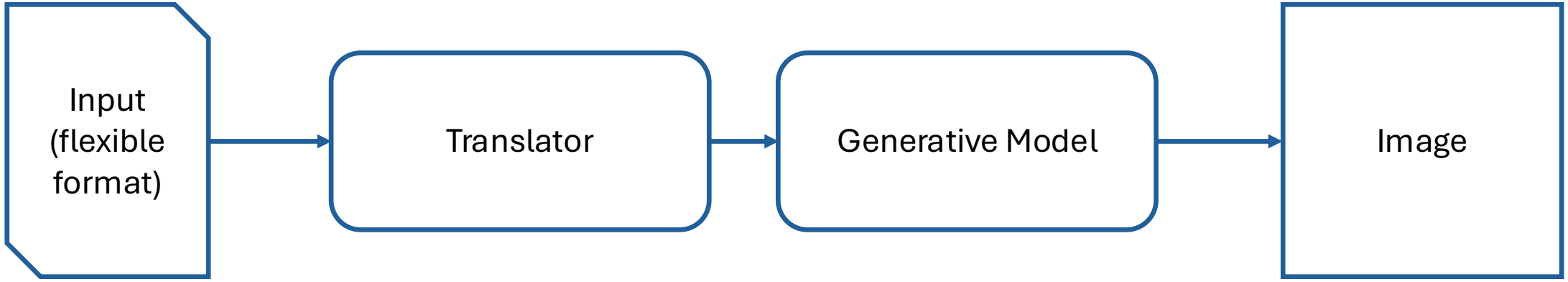
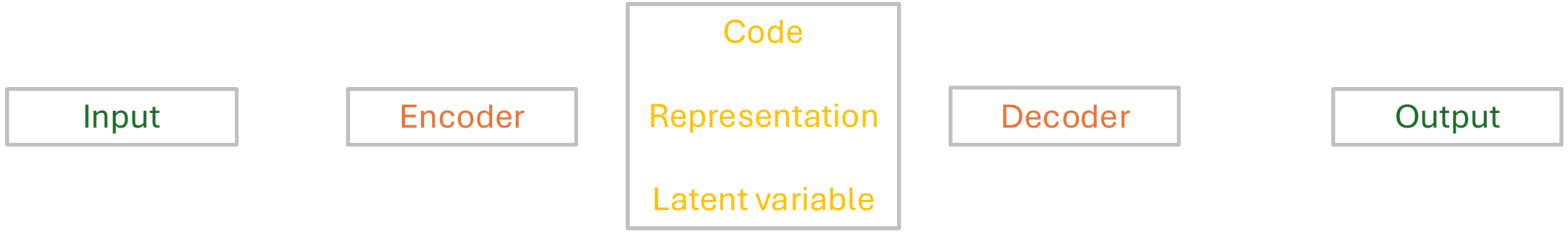
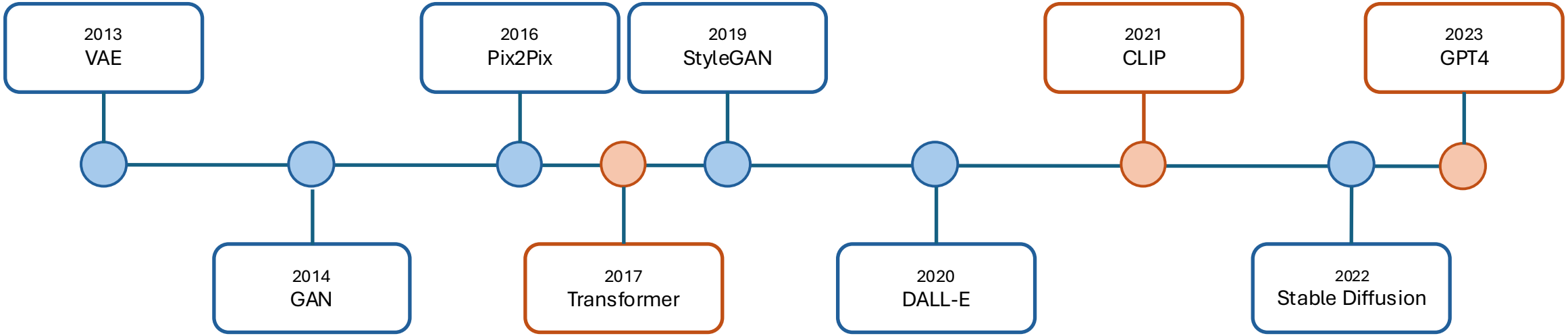
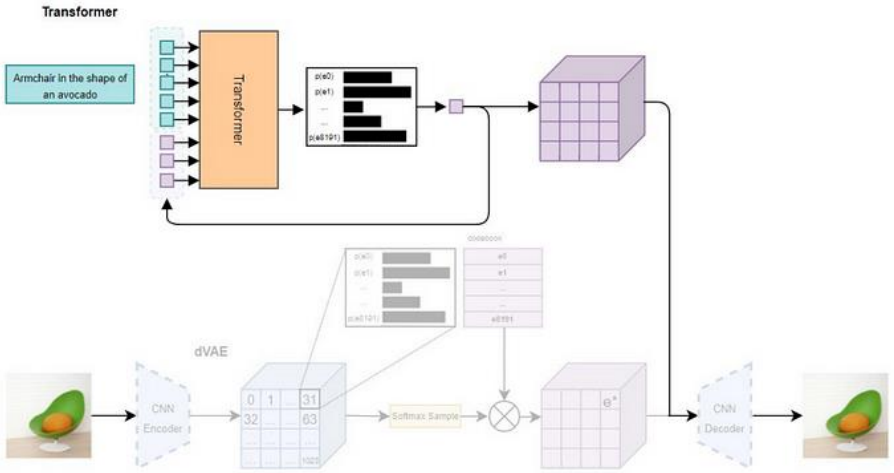




Timeline of Image Generation Models

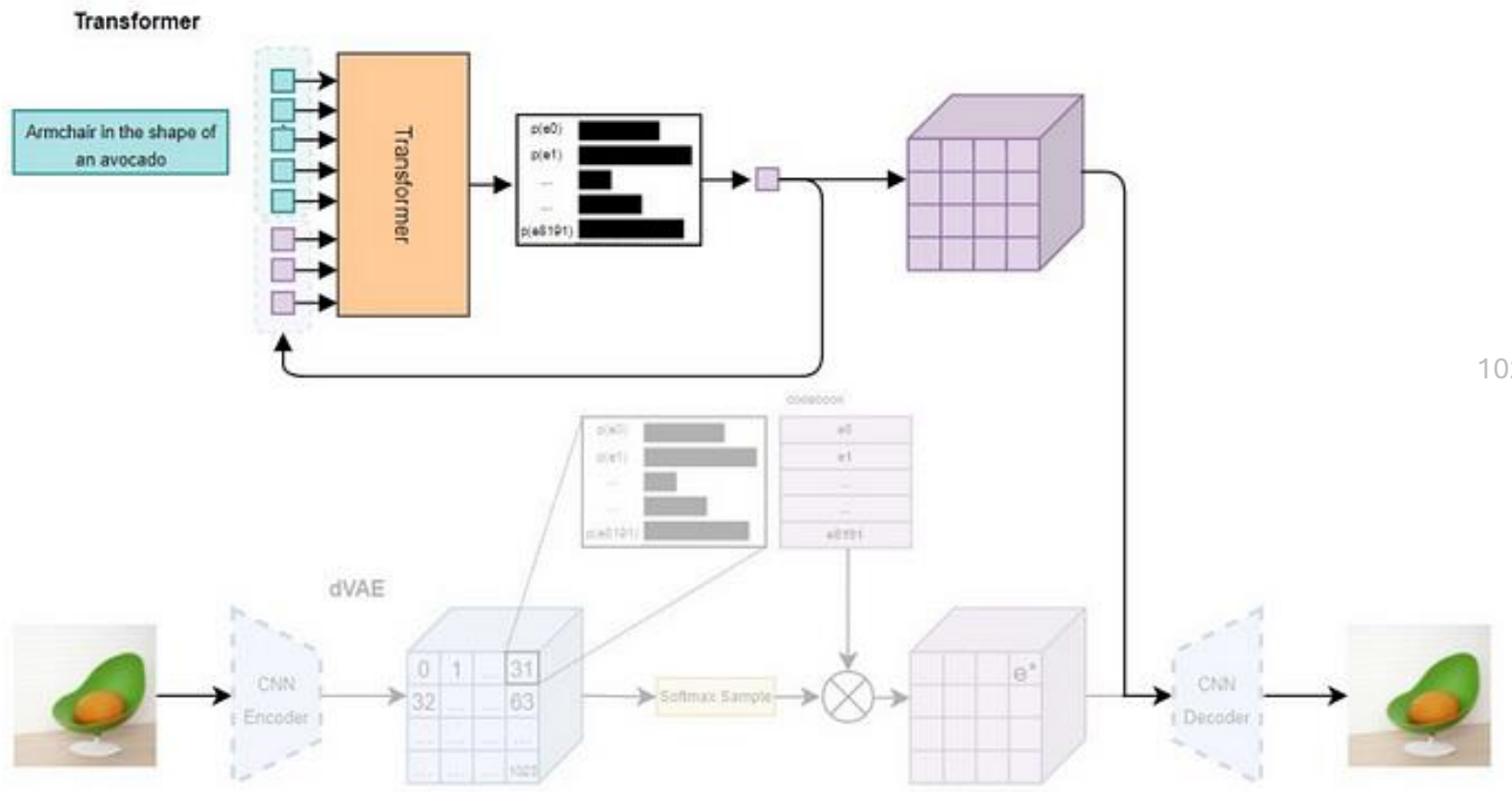




DALL-E: zero-shot text to image generation

Ramesh *et al*, 2020

Timeline of Image Generation Models

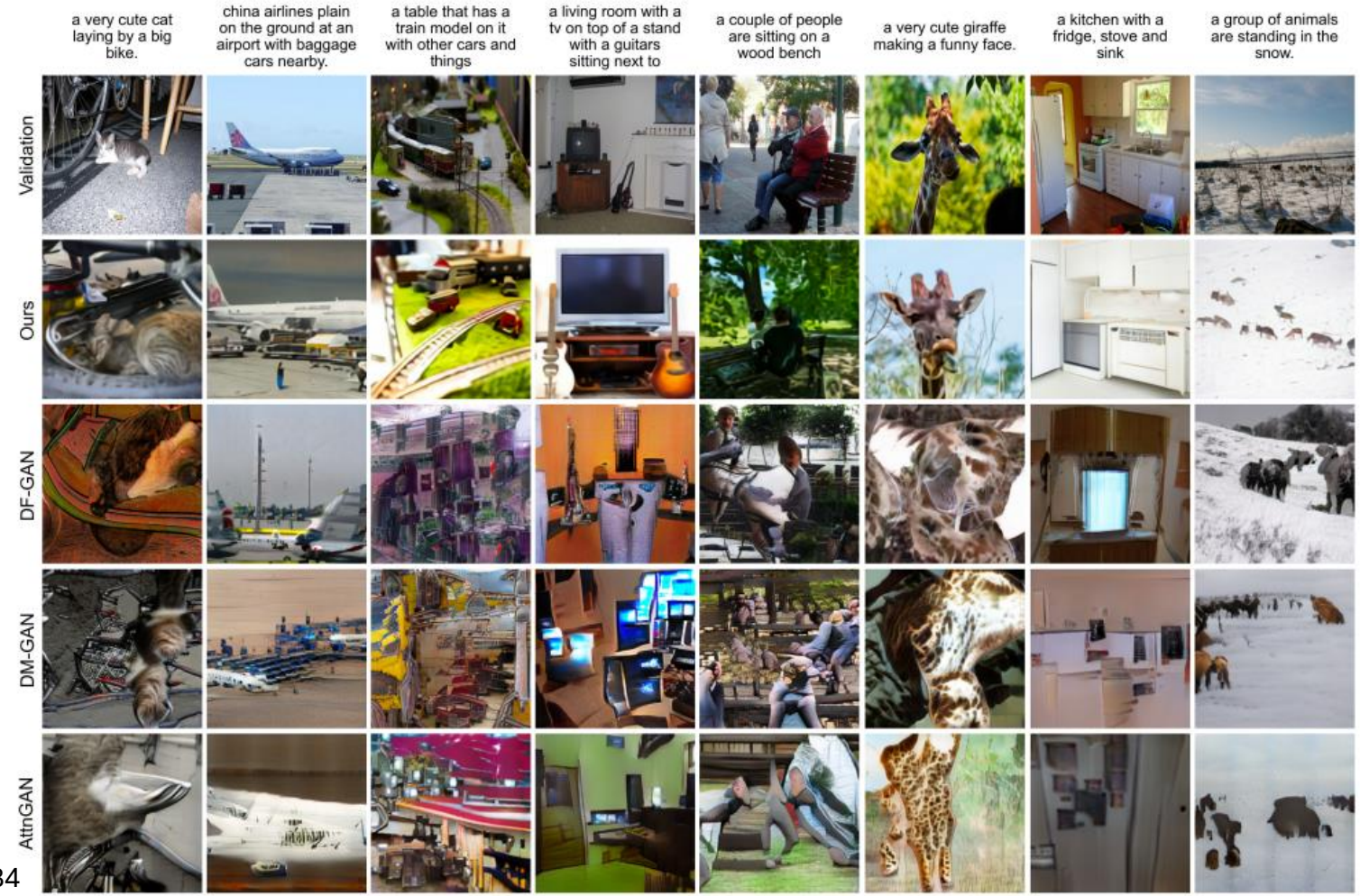


1024, 16 GB NVIDIA V100 GPUs

DALL-E: zero-shot text to image generation

Ramesh *et al*, 2020

Timeline of Image Generation Models



512x384

DALL-E: zero-shot text to image generation

Ramesh *et al*, 2020

Timeline of Image Generation Models

(a) a tapir made of accordion. a tapir with the texture of an accordion.

(b) an illustration of a baby hedgehog in a christmas sweater walking a dog

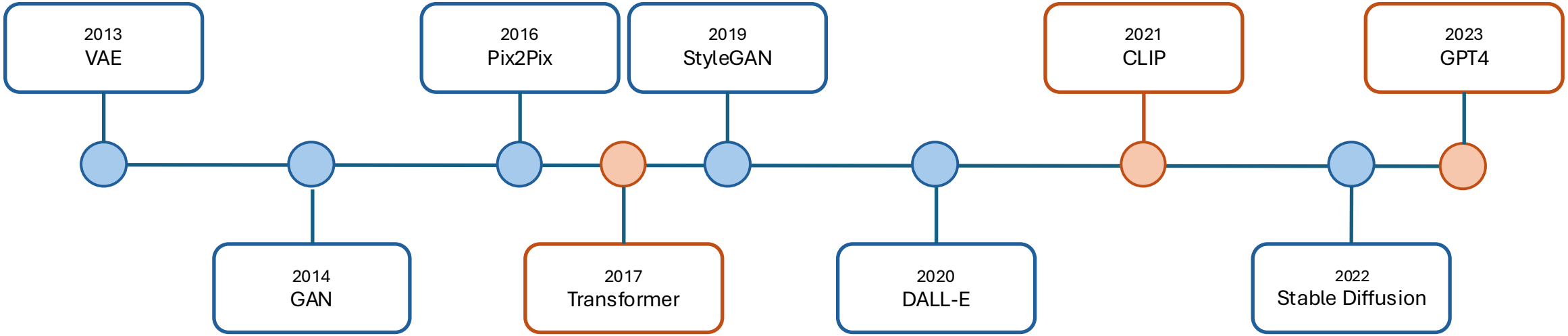
(c) a neon sign that reads "backprop". a neon sign that reads "backprop". backprop neon sign

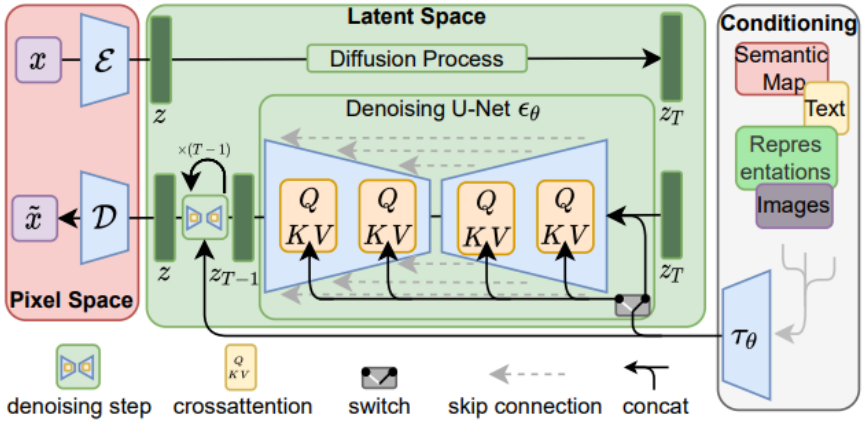
(d) the exact same cat on the top as a sketch on the bottom

DALL-E: zero-shot text to image generation

Ramesh *et al*, 2020

Timeline of Image Generation Models

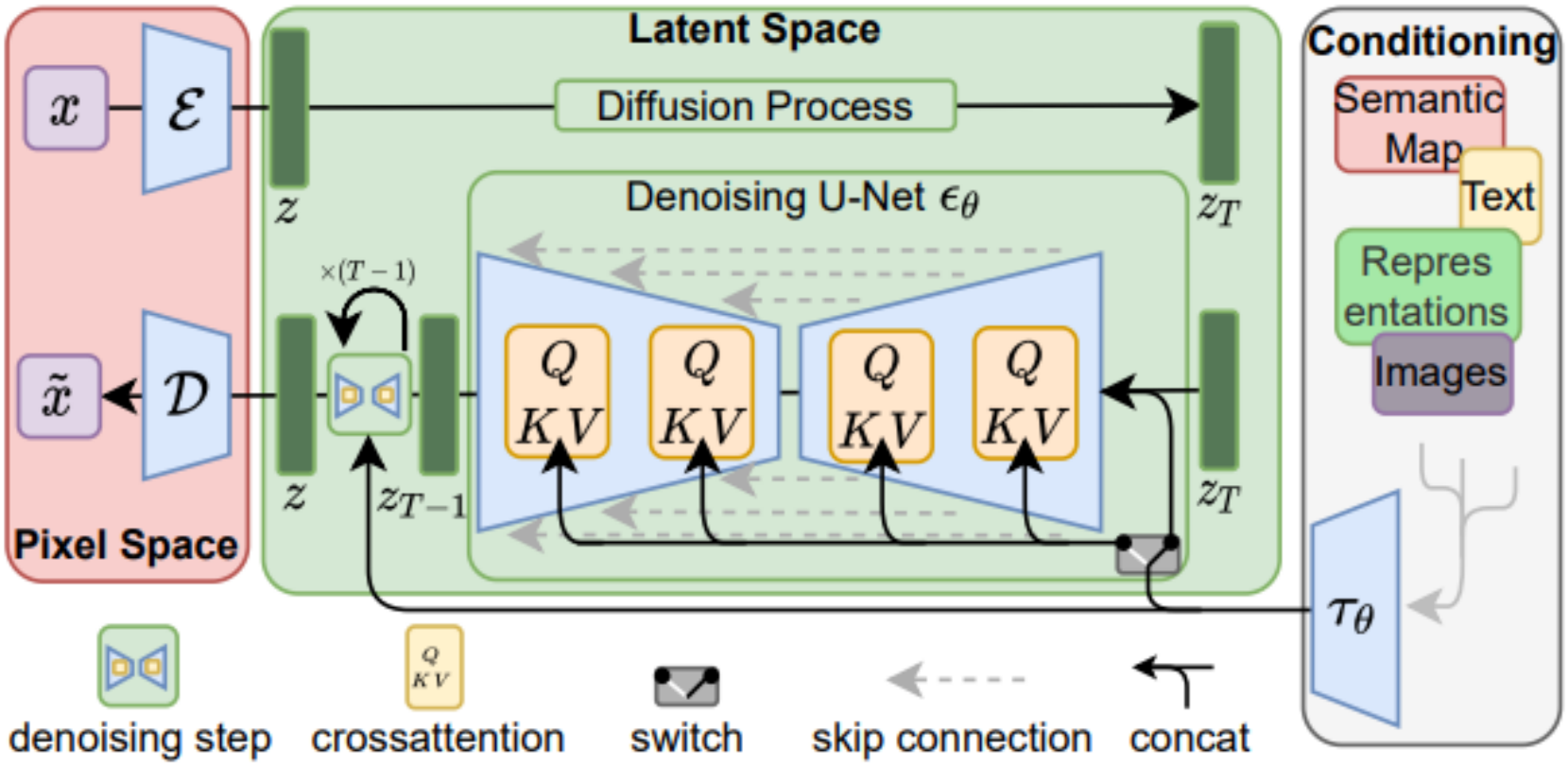




Latent Diffusion Models: high-resolution image generation

Rombach *et al*, 2022

Timeline of Image Generation Models



NVIDIA A100

Latent Diffusion Models: high-resolution image generation

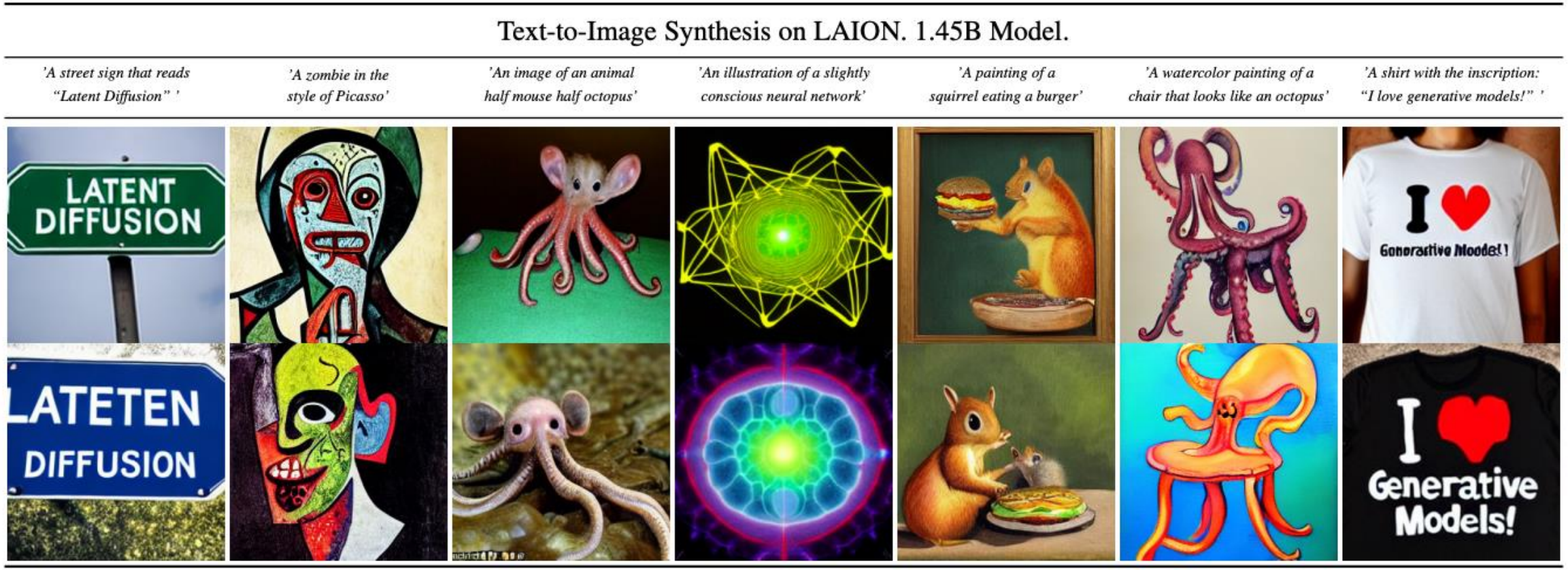
Rombach *et al*, 2022

Timeline of Image Generation Models

Latent Diffusion Models: high-resolution image generation

Rombach *et al*, 2022

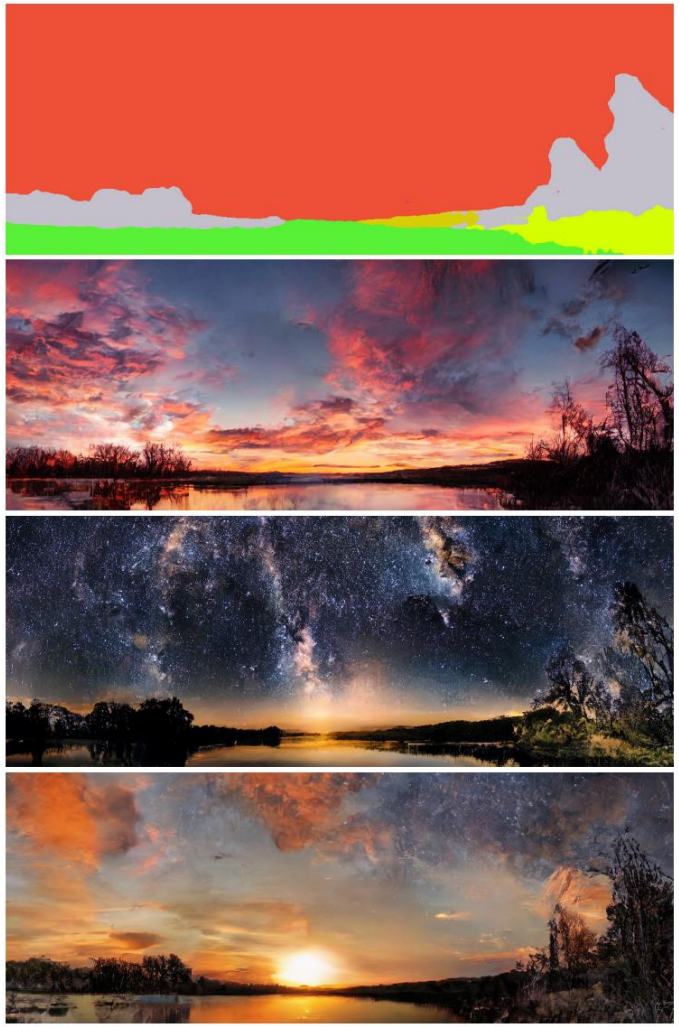
Timeline of Image Generation Models



Rombach *et al*, 2022

Latent Diffusion Models: high-resolution image generation

Timeline of Image Generation Models



1024x384

Latent Diffusion Models: high-resolution image generation

Rombach *et al*, 2022

Where were we in 2022?

- We can create rich representations from images
- We can generate random images from noise, other images and text
- We can finely control the generation of images using prompts
- We can generate high-resolution images

Where do we go from here?

- How can we extend this models to domain-specific images
- Can we tell in am image has been generated artificially

DALL-E and Latent Diffusion Model

Zero-Shot Text-to-Image Generation

Aditya Ramesh [1] Mikhail Pavlov [1] Gabriel Goh [1] Scott Gray [1]
Chelsea Voss [1] Alec Radford [1] Mark Chen [1] Ilya Sutskever [1]

Abstract

Text-to-image generation has traditionally focused on finding better modeling assumptions for training on a fixed dataset. These assumptions might involve complex architectures, auxiliary losses, or side information such as object part labels or segmentation masks supplied during training. We describe a simple approach for this task based on a transformer that autoregressively models the text and image tokens as a single stream of data. With sufficient data and scale, our approach is competitive with previous domain-specific models when evaluated in a zero-shot fashion.

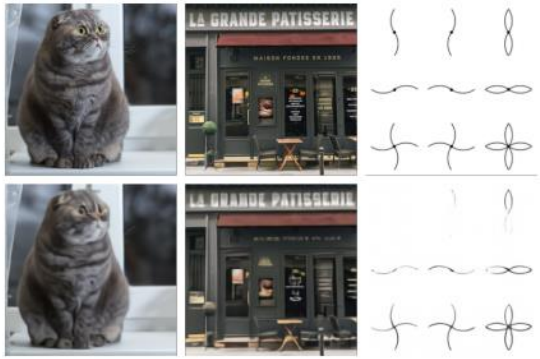


Figure 1. Comparison of original images (top) and reconstructions from the discrete VAE (bottom). The encoder downsamples the spatial resolution by a factor of 8. While details (e.g., the texture of the cat's fur, the writing on the storefront, and the thin lines in the illustration) are sometimes lost or distorted, the main features of the image are still typically recognizable. We use a large vocabulary size of 8192 to mitigate the loss of information.

1. Introduction

Modern machine learning approaches to text to image synthesis started with the work of Mansimov et al. (2015), who showed that the DRAW Gregor et al. (2015) generative model, when extended to condition on image captions, could also generate novel visual scenes. Reed et al. (2016b) later demonstrated that using a generative adversarial network (Goodfellow et al., 2014), rather than a recurrent variational auto-encoder, improved image fidelity. Reed et al. (2016b) showed that this system could not only generate objects with recognizable properties, but also could *zero-shot* generalize to held-out categories.

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High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach[1] * Andreas Blattmann[1] * Dominik Lorenz[1] Patrick Esser[R] Björn Ommer[1]
[1]Ludwig Maximilian University of Munich & IWR, Heidelberg University, Germany [R]Runway ML
https://github.com/CompVis/latent-diffusion

Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

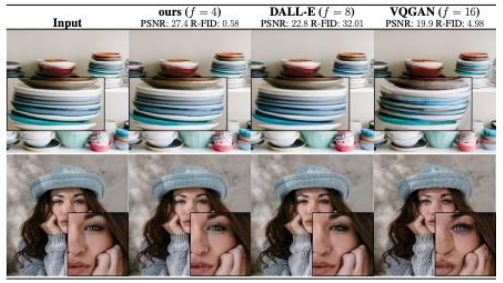

Figure 1. Boosting the upper bound on achievable quality with less agressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512^2 px. We denote the spatial downsampling factor by f. Reconstruction FIDs [29] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

results in image synthesis [30,85] and beyond [7,45,48,57], and define the state-of-the-art in class-conditional image synthesis [15,31] and super-resolution [72]. Moreover, even unconditional DMs can readily be applied to tasks such as inpainting and colorization [85] or stroke-based synthesis [53], in contrast to other types of generative models [19,46,69]. Being likelihood-based models, they do not exhibit mode-collapse and training instabilities as GANs and, by heavily exploiting parameter sharing, they can

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DALL-E and Latent Diffusion Model

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Aditya Ramesh [1] Mikhail Pavlov [1] Gabriel Goh [1] Scott Gray [1]
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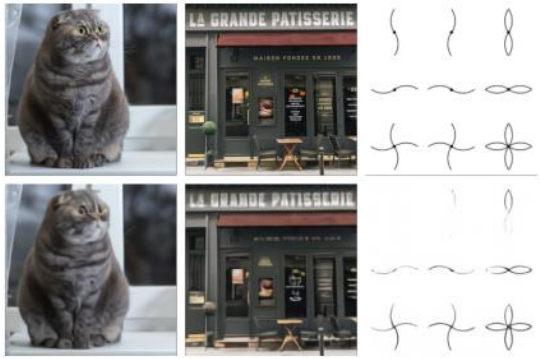

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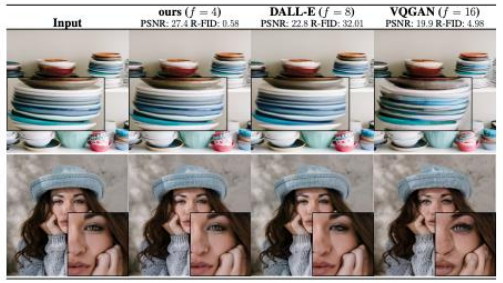

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DALL-E: Zero-Shot Text to Image Generation

Objectives:

- Create model to generate images from text descriptions

- Want the model to generalize to image/text not used for training (zero-shot)

Ramesh *et al*, 2020

DALL-E: Zero-Shot Text to Image Generation

Basic Idea:

- Create a model that can create images from representations
- Generate representations from text
- Align image and text representations so:

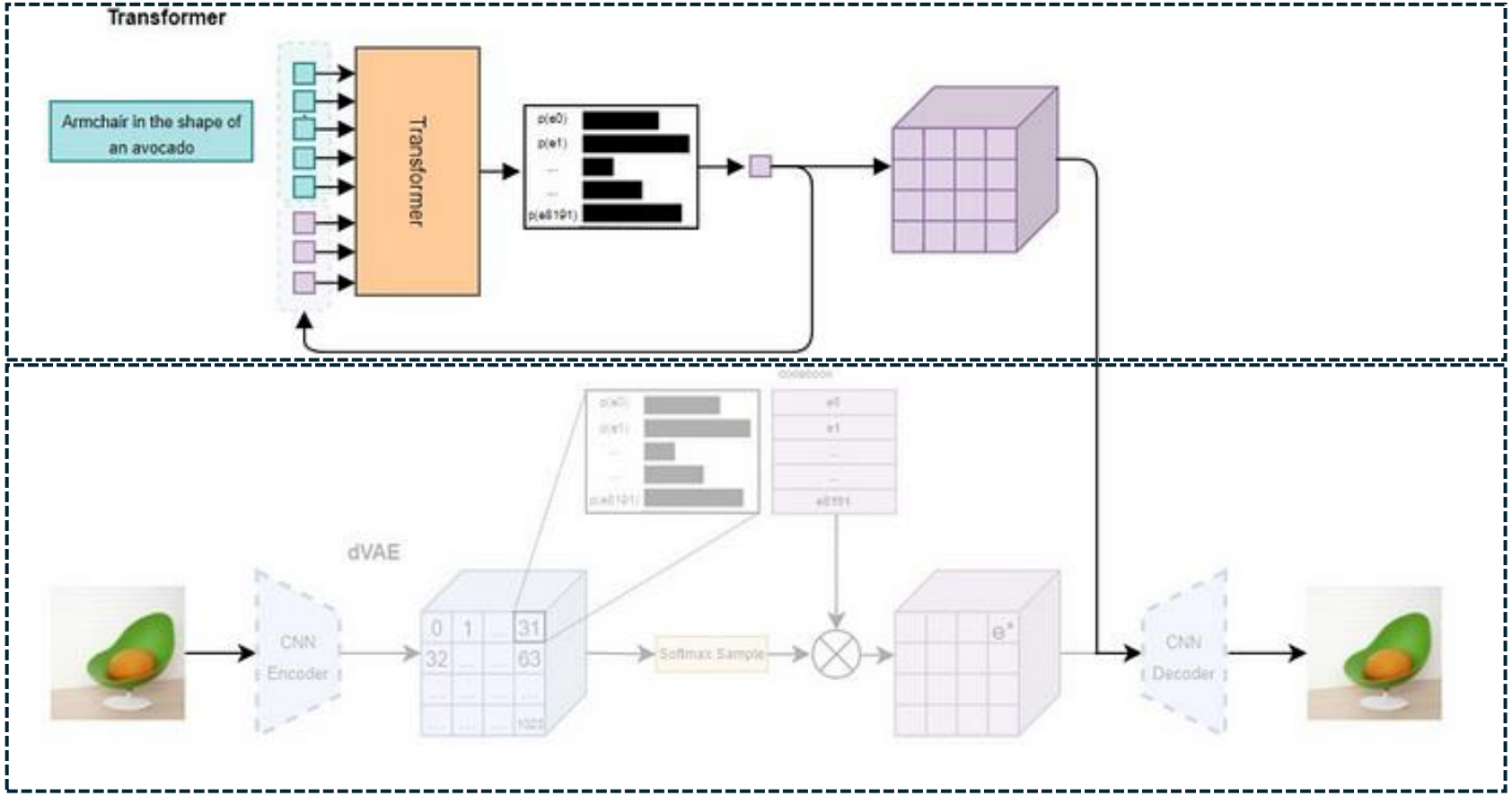
Text ⟹ Representation ⟹ Image

Ramesh *et al*, 2020

DALL-E: Zero-Shot Text to Image Generation

Text Encoder
(Stage 2)

Autoencoder
(Stage 1)



Ramesh *et al*, 2020

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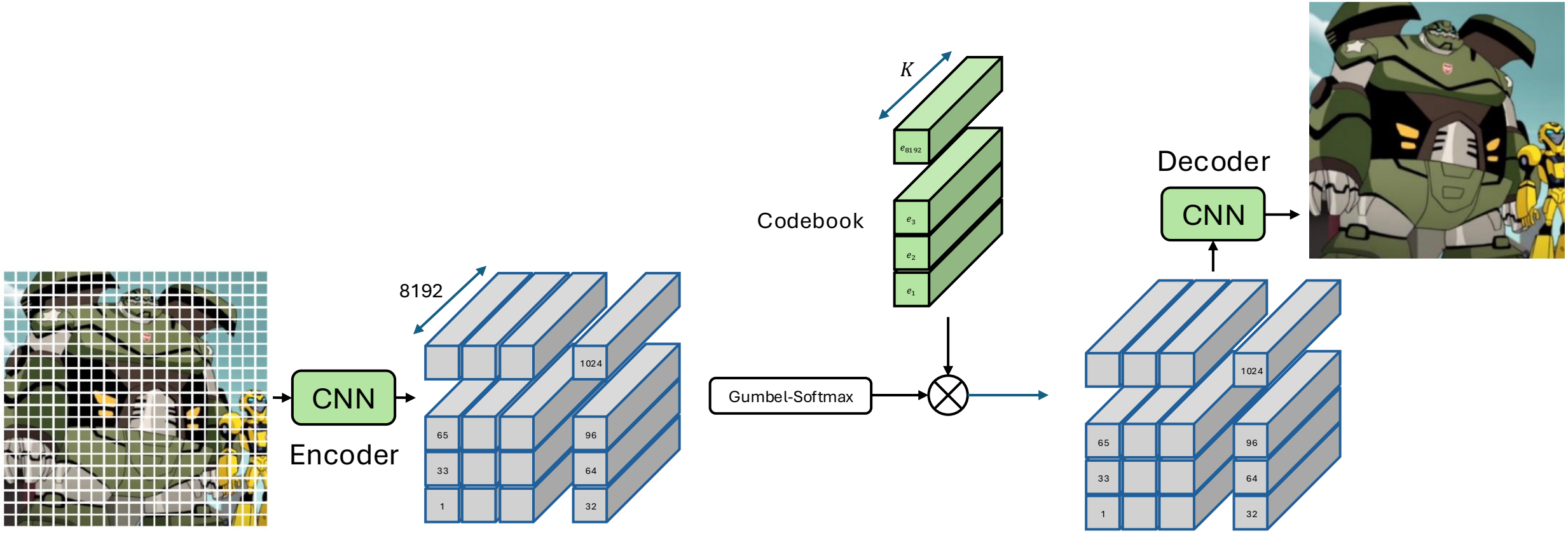


Divide image into patches (32x32)

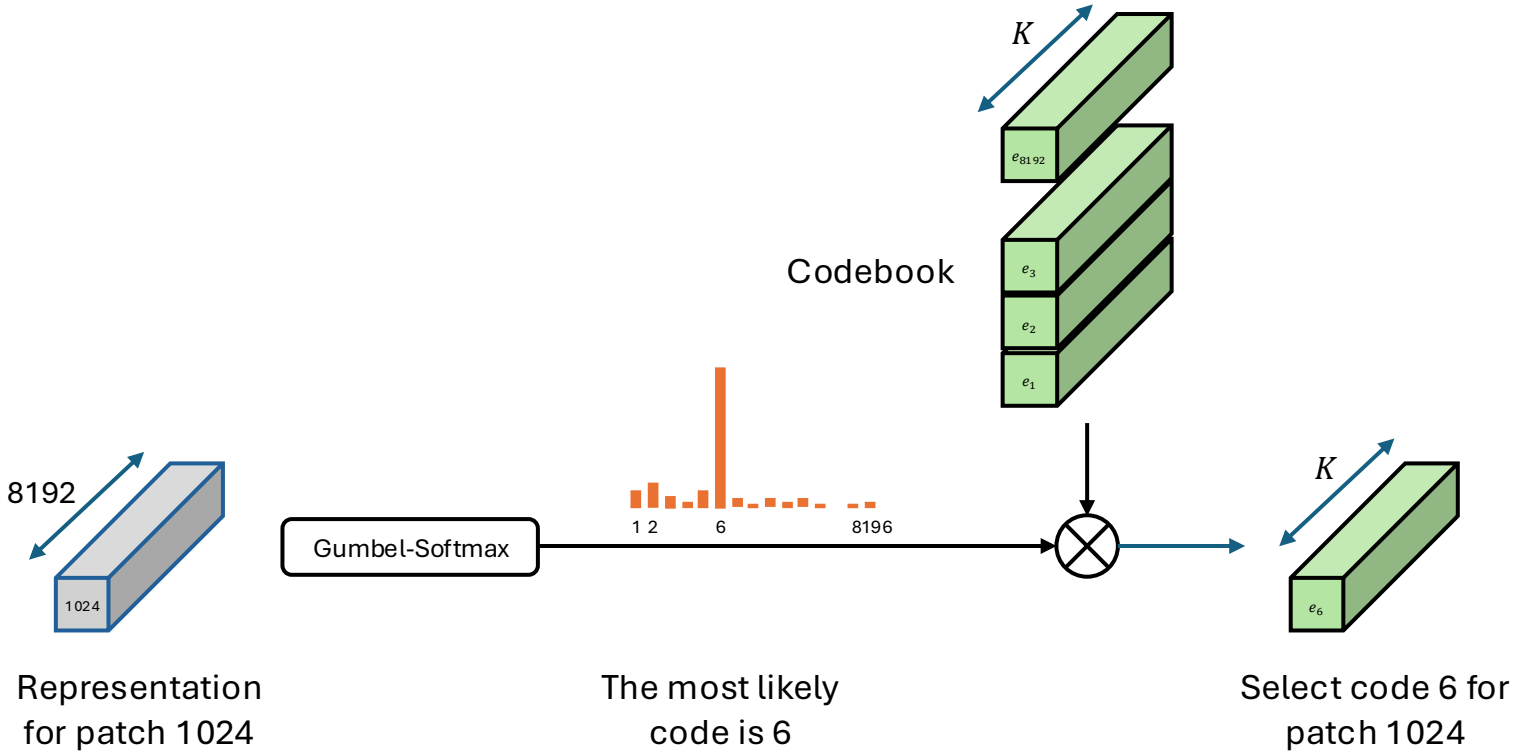
Each patch is represented by a vector of length 8192

For each patch select the most likely element of the vector

Fill the representation of each patch with the code selected by Gumbel-Softmax

Ramesh *et al*, 2020

DALL-E: Zero-Shot Text to Image Generation



Ramesh *et al*, 2020

DALL-E: Zero-Shot Text to Image Generation

Learning objective:

Image

Representation

$$\log p_\theta(x) \geq \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z) - \beta D_{KL}(q_\phi(z|x)|p(z))]$$

Model parameters

- $p_\theta(x|z)$: model that generates image x from representation z
- $q_\phi(z|x)$: model that generates representation z from image x
- $p(z)$: prior distribution for representation z

DALL-E: Zero-Shot Text to Image Generation

Learning objective (dVAE):

Image

Representation

$$\log p_\theta(x) \geq \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z) - \beta D_{KL}(q_\phi(z|x)|p(z))]$$

Model parameters

- p_θ:$(x|z)$ CNN that generates image x from representation z
- $q_\phi(z|x)$: CNN that generates representation z from image x
- $p(z)$: prior uniform distribution for representation z

$$z \sim q_\phi(z|x): z = f(z'; C), z' = g(x, \epsilon; W), \epsilon \sim \mathcal{N}(0,1)$$

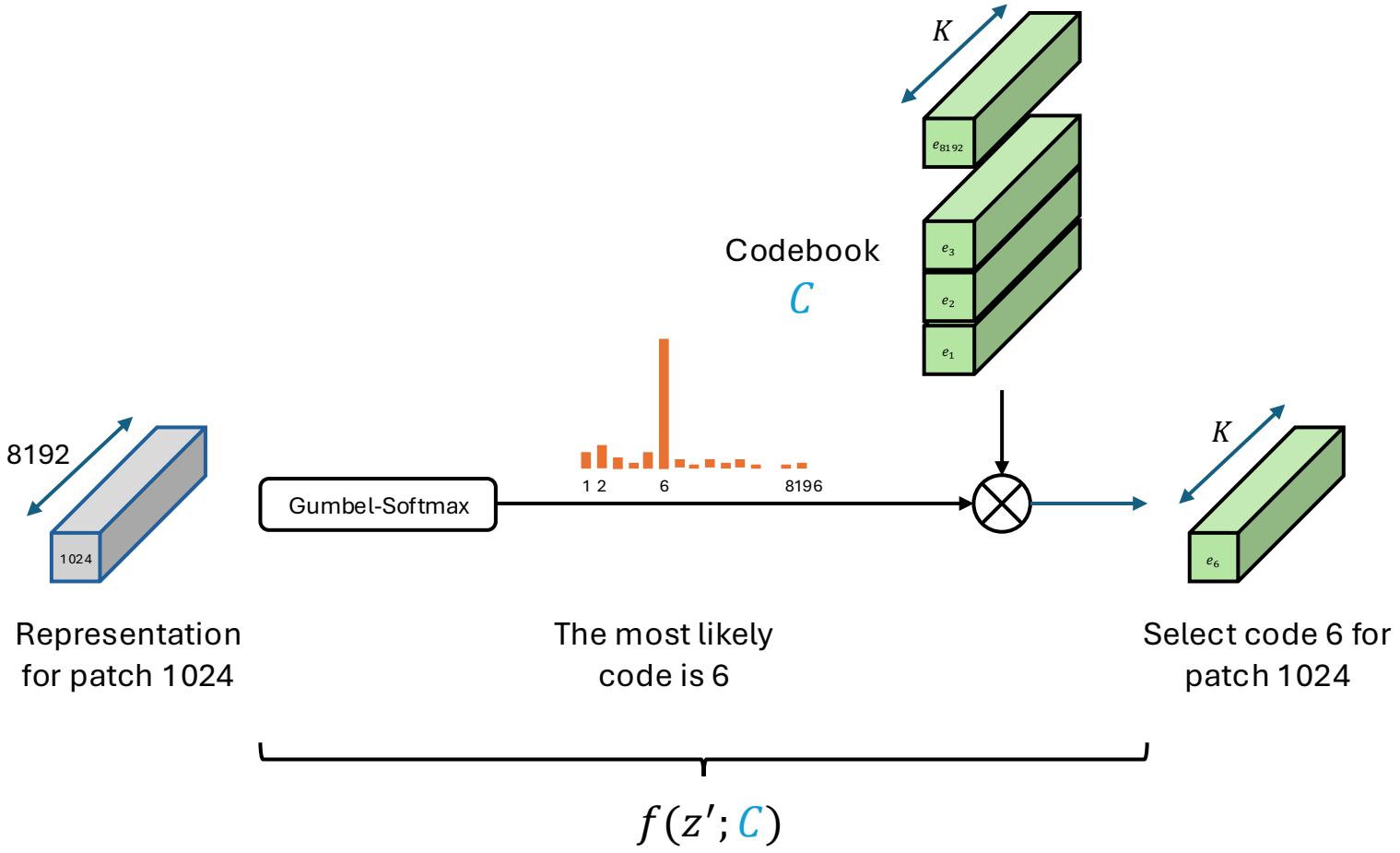
DALL-E: Zero-Shot Text to Image Generation

Learning objective (dVAE):

$$z \sim q_{\phi}(z|x): z = f(z'; C), z' = g(x, \epsilon; W), \epsilon \sim \mathcal{N}(0,1)$$

- ϵ: sample from a (standard) Gaussian distribution
- $g(x, \epsilon; W)$: CNN encoder with inputs x and ϵ
- $f(z'; C)$: function that selects from C according to z'

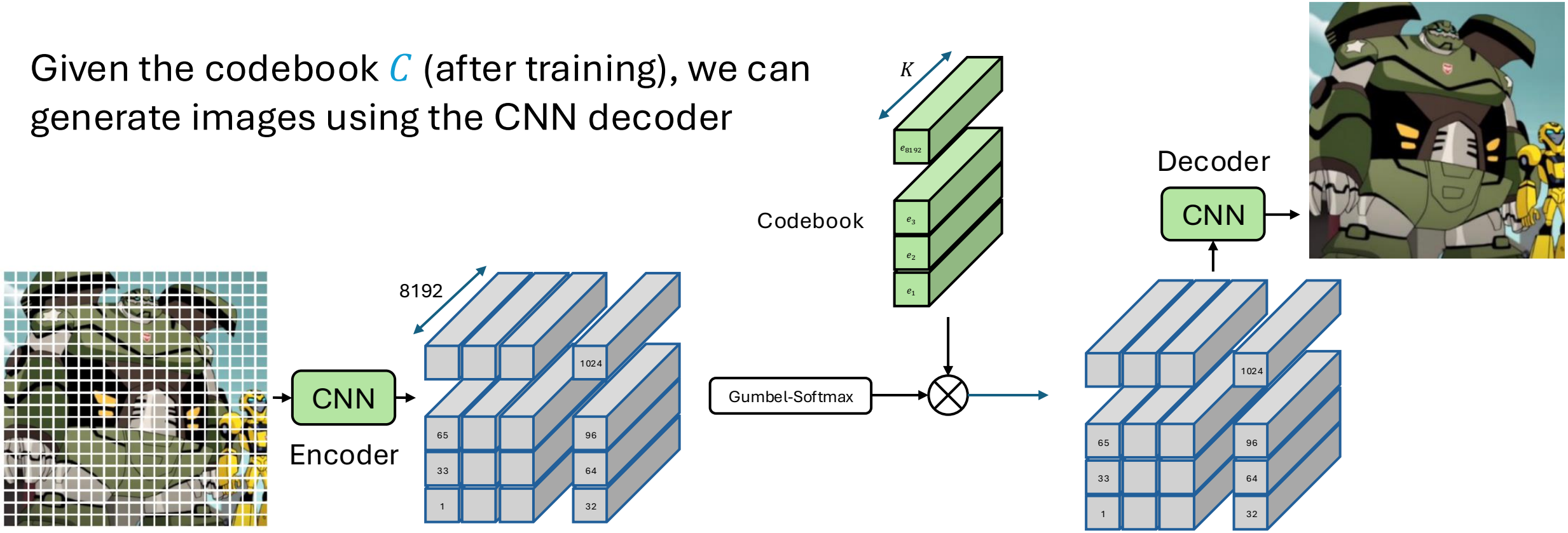
DALL-E: Zero-Shot Text to Image Generation



Ramesh *et al*, 2020

DALL-E: Zero-Shot Text to Image Generation

Given the codebook C (after training), we can generate images using the CNN decoder



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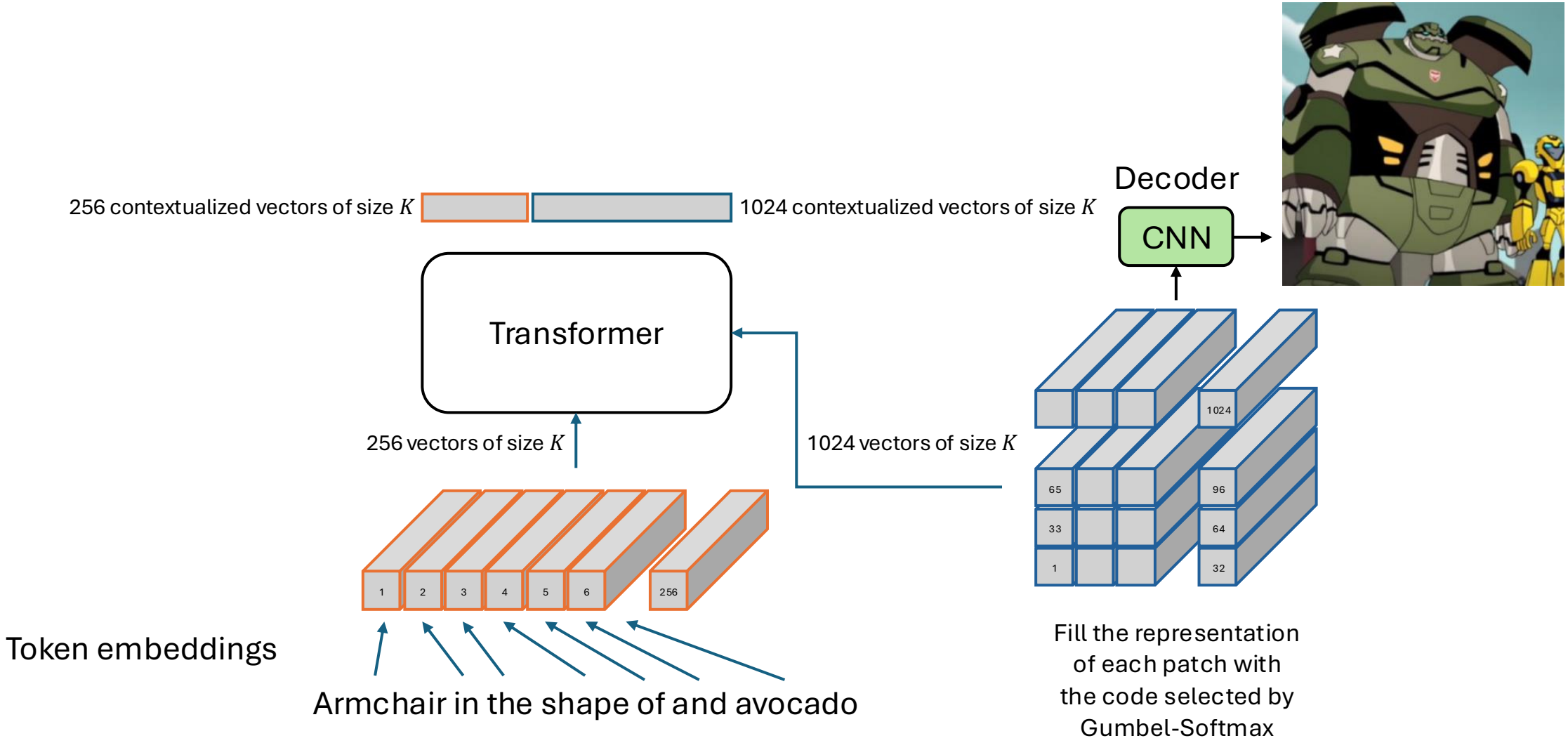

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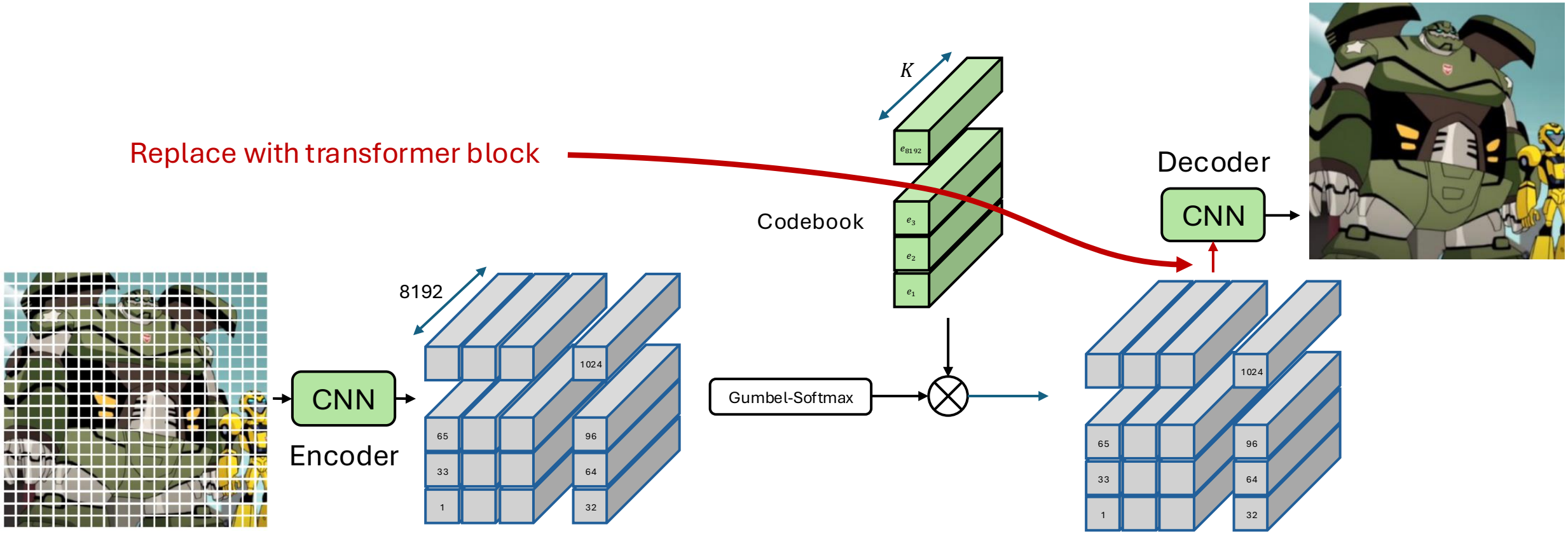

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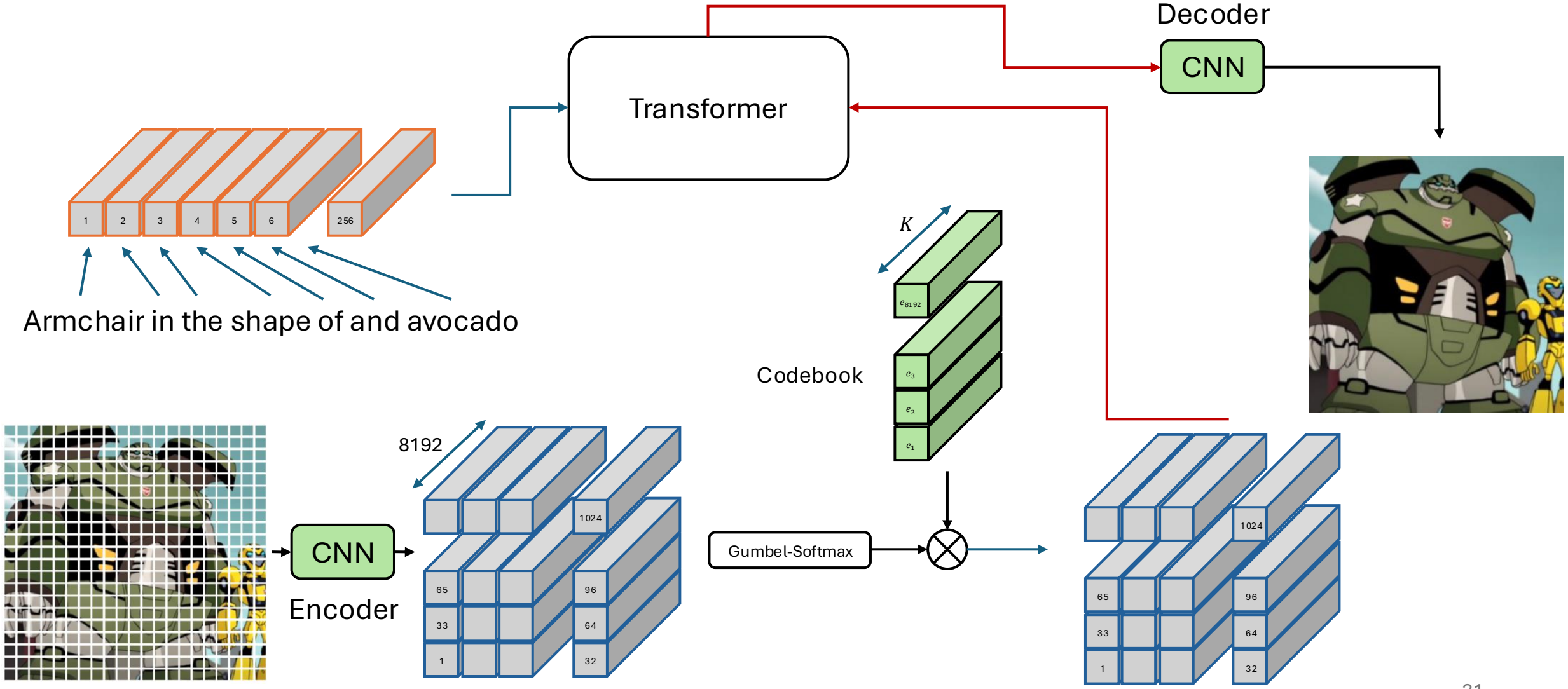


Ramesh *et al*, 2020

DALL-E: Zero-Shot Text to Image Generation

Learning objective (dVAE):

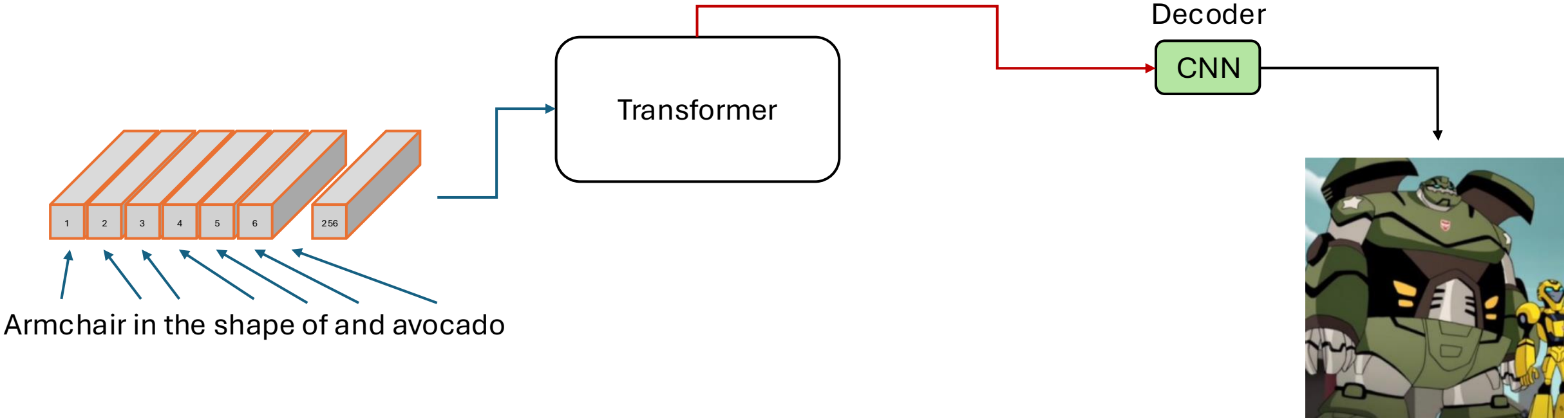
Image Representation Text description

$$\log p_\theta(x) \geq \mathbb{E}_{y,z\sim q_\phi(y,z|x)}[\log p_\theta(x|z,y) - \beta D_{KL}(q_\phi(y,z|x)|p(y,z))]$$

Model parameters

- $p_\theta(x|z,y)$: CNN that generates x from z and y
- $q_\phi(y,z|x)$: CNN that generates y and z from x
- $p(y,z)$: prior uniform distribution for z and y

DALL-E: Zero-Shot Text to Image Generation



After training, we can generate images using the CNN decoder and the transformer to encode the text description

Ramesh *et al*, 2020

Recap

- DALL-E is a combination of two building blocks
  - Discrete VAE to reconstruct images from representations
  - Transformer to represent image descriptions
- The model is trained end-to-end using a VAE objective
- At inference, we encode text and decode an image

Transformer CNN

DALL-E and Latent Diffusion Model

Zero-Shot Text-to-Image Generation

Aditya Ramesh [1] Mikhail Pavlov [1] Gabriel Goh [1] Scott Gray [1]
Chelsea Voss [1] Alec Radford [1] Mark Chen [1] Ilya Sutskever [1]

Abstract

Text-to-image generation has traditionally focused on finding better modeling assumptions for training on a fixed dataset. These assumptions might involve complex architectures, auxiliary losses, or side information such as object part labels or segmentation masks supplied during training. We describe a simple approach for this task based on a transformer that autoregressively models the text and image tokens as a single stream of data. With sufficient data and scale, our approach is competitive with previous domain-specific models when evaluated in a zero-shot fashion.

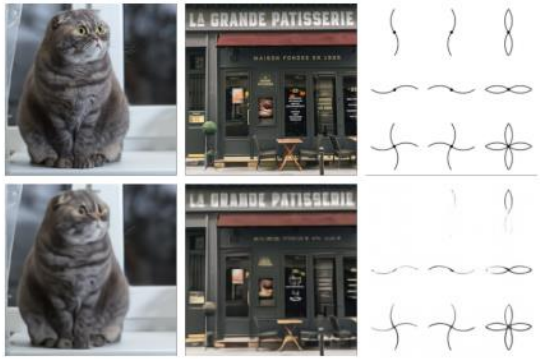


Figure 1. Comparison of original images (top) and reconstructions from the discrete VAE (bottom). The encoder downsamples the spatial resolution by a factor of 8. While details (e.g., the texture of the cat's fur, the writing on the storefront, and the thin lines in the illustration) are sometimes lost or distorted, the main features of the image are still typically recognizable. We use a large vocabulary size of 8192 to mitigate the loss of information.

1. Introduction

Modern machine learning approaches to text to image synthesis started with the work of Mansimov et al. (2015), who showed that the DRAW Gregor et al. (2015) generative model, when extended to condition on image captions, could also generate novel visual scenes. Reed et al. (2016b) later demonstrated that using a generative adversarial network (Goodfellow et al., 2014), rather than a recurrent variational auto-encoder, improved image fidelity. Reed et al. (2016b) showed that this system could not only generate objects with recognizable properties, but also could *zero-shot* generalize to held-out categories.

tioning model pretrained on MS-COCO. More recently, Cho et al. (2020) also propose a method that involves optimizing the input to a pretrained cross-modal masked language model. While significant increases in visual fidelity have occurred as a result of the work since Mansimov et al. (2015), samples can still suffer from severe artifacts such as object

Zero-shot text-to-image generation
A Ramesh, M Pavlov, G Goh, S Gray… - International …, 2021 - proceedings.mlr.press
Text-to-image generation has traditionally focused on finding better modeling assumptions for training on a fixed dataset. These assumptions might involve complex architectures …
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High-Resolution Image Synthesis with Latent Diffusion Models

Robin Rombach[1] * Andreas Blattmann[1] * Dominik Lorenz[1] Patrick Esser[R] Björn Ommer[1]
[1]Ludwig Maximilian University of Munich & IWR, Heidelberg University, Germany [R]Runway ML
https://github.com/CompVis/latent-diffusion

Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state-of-the-art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including text-to-image synthesis, unconditional image generation and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

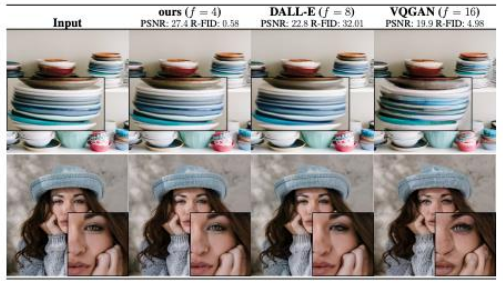

Figure 1. Boosting the upper bound on achievable quality with less agressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512^2 px. We denote the spatial downsampling factor by f. Reconstruction FIDs [29] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

results in image synthesis [30,85] and beyond [7,45,48,57], and define the state-of-the-art in class-conditional image synthesis [15,31] and super-resolution [72]. Moreover, even unconditional DMs can readily be applied to tasks such as inpainting and colorization [85] or stroke-based synthesis [53], in contrast to other types of generative models [19,46,69]. Being likelihood-based models, they do not exhibit mode-collapse and training instabilities as GANs and, by heavily exploiting parameter sharing, they can

High-resolution image synthesis with **latent diffusion models**
R Rombach, A Blattmann, D Lorenz… - Proceedings of the …, 2022 - openaccess.thecvf.com
… their **quality** and flexibility, we apply them in the **latent** space of … In contrast to previous work, training **diffusion models** on such a … into the **model** architecture, we turn **diffusion models** into …
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Latent Diffusion Model

Objectives:

- Create model to generate images from text descriptions

- Want the model to generalize to image/text not used for training

- Want the model to produce high-resolution images

Rombach *et al*, 2022

Latent Diffusion Model

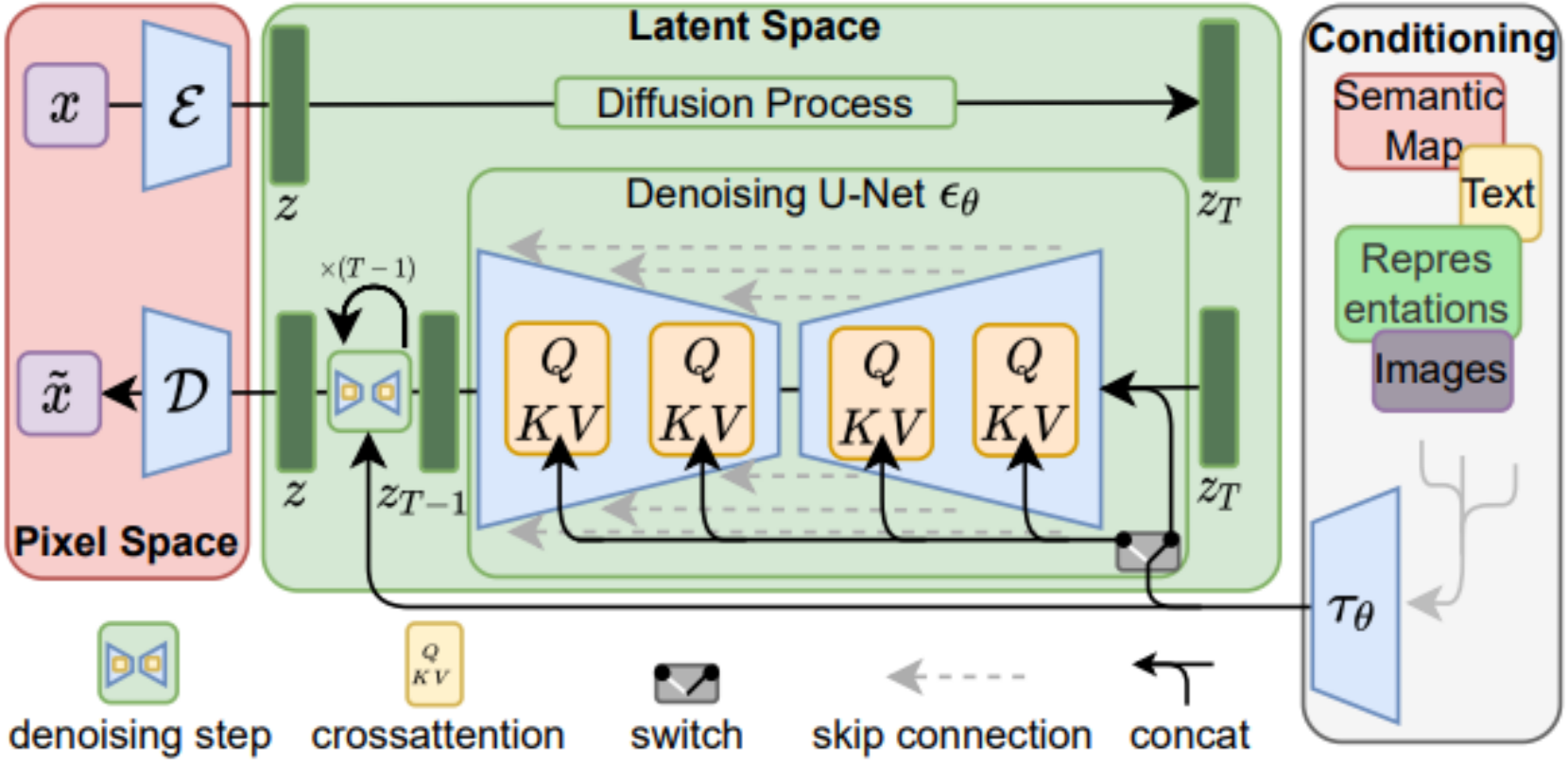
Basic Idea (same as in DALL-E):

- Create a model that can create images from representations
- Generate representations from text
- Align image and text representations so:

Text ⟹ Representation ⟹ Image

Rombach *et al*, 2022

Latent Diffusion Model

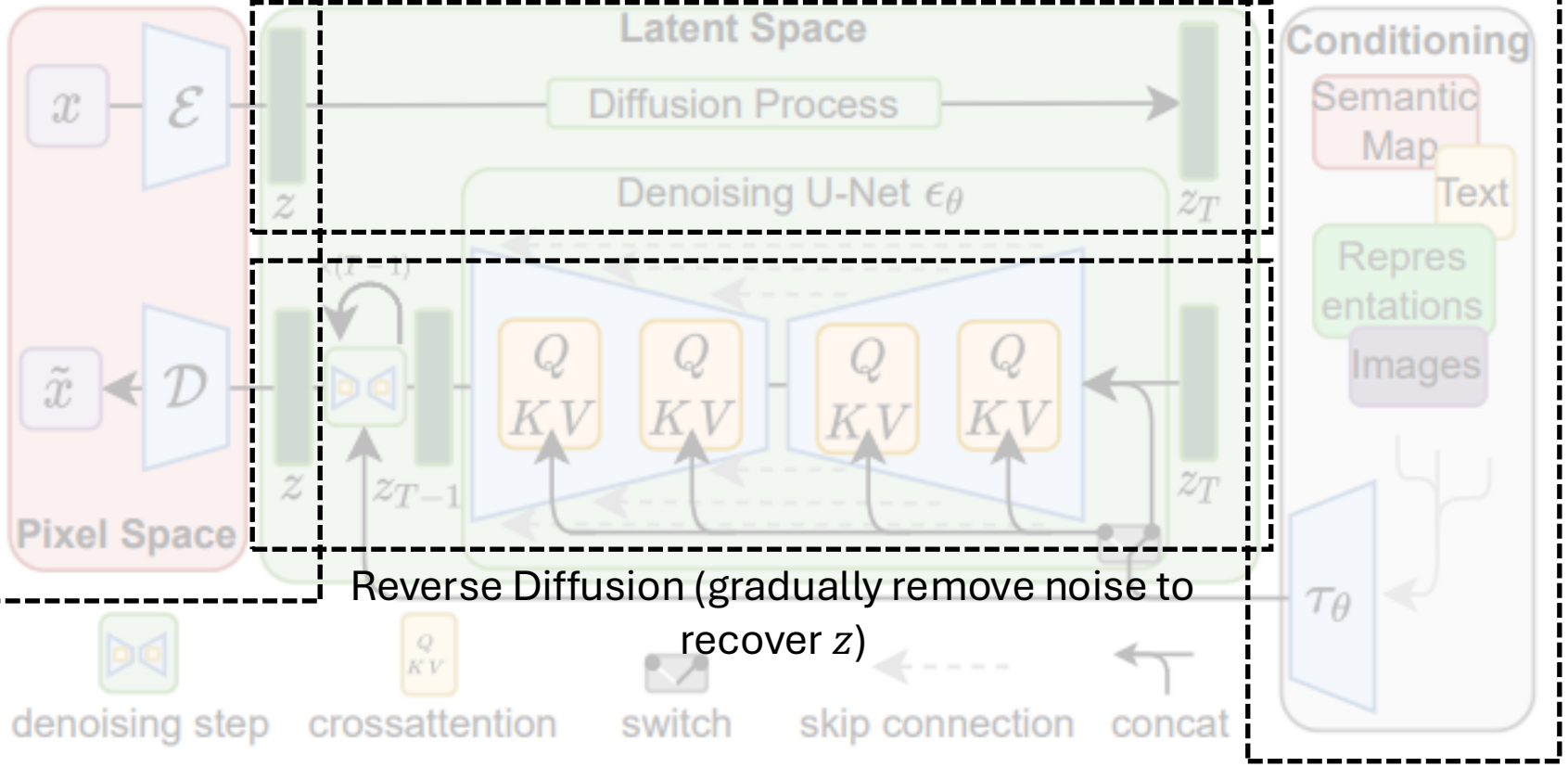


Rombach *et al*, 2022

Latent Diffusion Model

Forward Diffusion (gradually add noise to z)

Autoencoder



Reverse Diffusion (gradually remove noise to recover z)

Represent Conditioning Data

Rombach *et al*, 2022

Latent Diffusion Model

Forward Diffusion (gradually add noise to z)

Song *et al*, 2021

z z_2 z_3 z_T

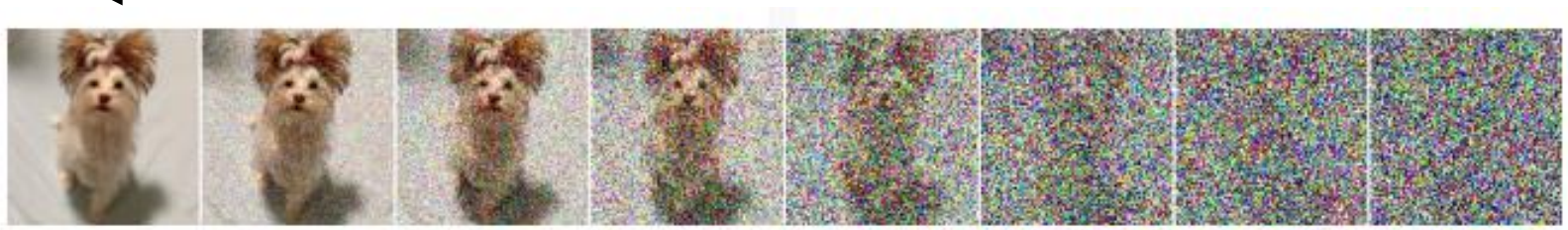
$$z_i = \beta z_{i-1} + (1-\beta)\epsilon_i$$

$$\epsilon_i \sim \mathcal{N}(0,1)$$

Rombach *et al*, 2022

Latent Diffusion Model

Reverse Diffusion (gradually remove noise to recover z)



Song *et al*, 2021

z z_2 z_3 z_T

Gaussian noise

$$L_{DM} = \mathbb{E}_{x,t,\epsilon}[\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2]$$

Difference between representation at time t and reconstruction from $t-1$, *i.e.*, z_{t-1}

$$\epsilon_\theta(z_t, t) = (z_t - x_\theta(z_{t-1}, t))$$

Model that predicts z_t from z_{t-1}

Rombach *et al*, 2022

Latent Diffusion Model

Reverse Diffusion (gradually remove noise to recover z)

Song *et al*, 2021

z z_2 z_3 z_{T-1} z_T

$\tilde{z}_{T-1}$

Predict

$$z_{T-1} - \tilde{z}_{T-1} \sim \mathcal{N}(0,1)$$

We want to make predictions from T to $t = 2$ so the difference between predictions and the noisy images created in the forward diffusion looks like Gaussian noise

Rombach *et al*, 2022

Latent Diffusion Model

Reverse Diffusion (gradually remove noise to recover z)

Song *et al*, 2021

z z_2 z_3 z_{T-1} z_T

$\tilde{z}_{T-1}$

$$z_{T-1} - \tilde{z}_{T-1} \sim \mathcal{N}(0,1)$$

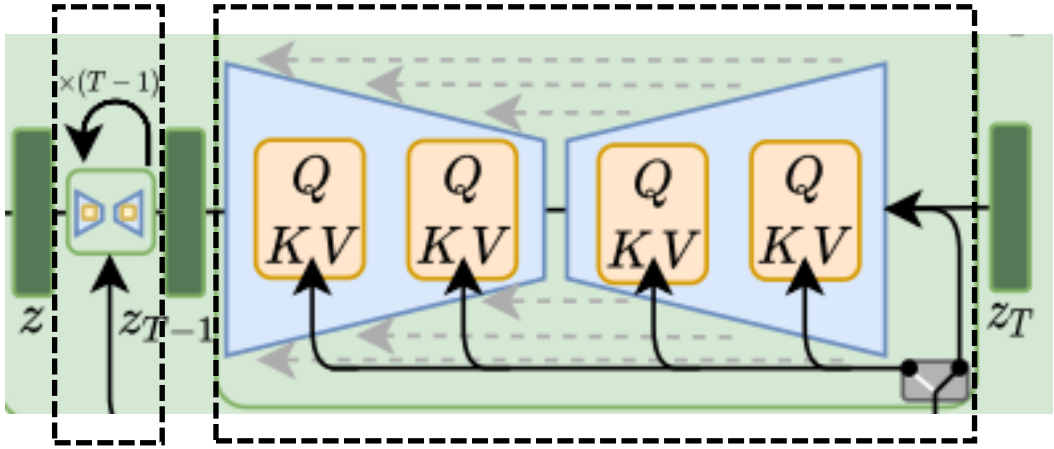
Predict

Note: unlike GANs, we do not go straight from noise to image, we do it in T steps

Rombach *et al*, 2022

Latent Diffusion Model

$x_\theta(z_2, 1)$ $\cdots$ $x_\theta(z_{T-1}, T-2)$ $x_\theta(z_T, T-1)$



Attention UNet

Gaussian noise

$$L_{DM} = \mathbb{E}_{x,t,\epsilon}[\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2]$$

Difference between representation at time t and reconstruction from $t+1$, *i.e.*, z_{t+1}

$$\epsilon_\theta(z_t, t) = (z_t - x_\theta(z_{t+1}, t))$$

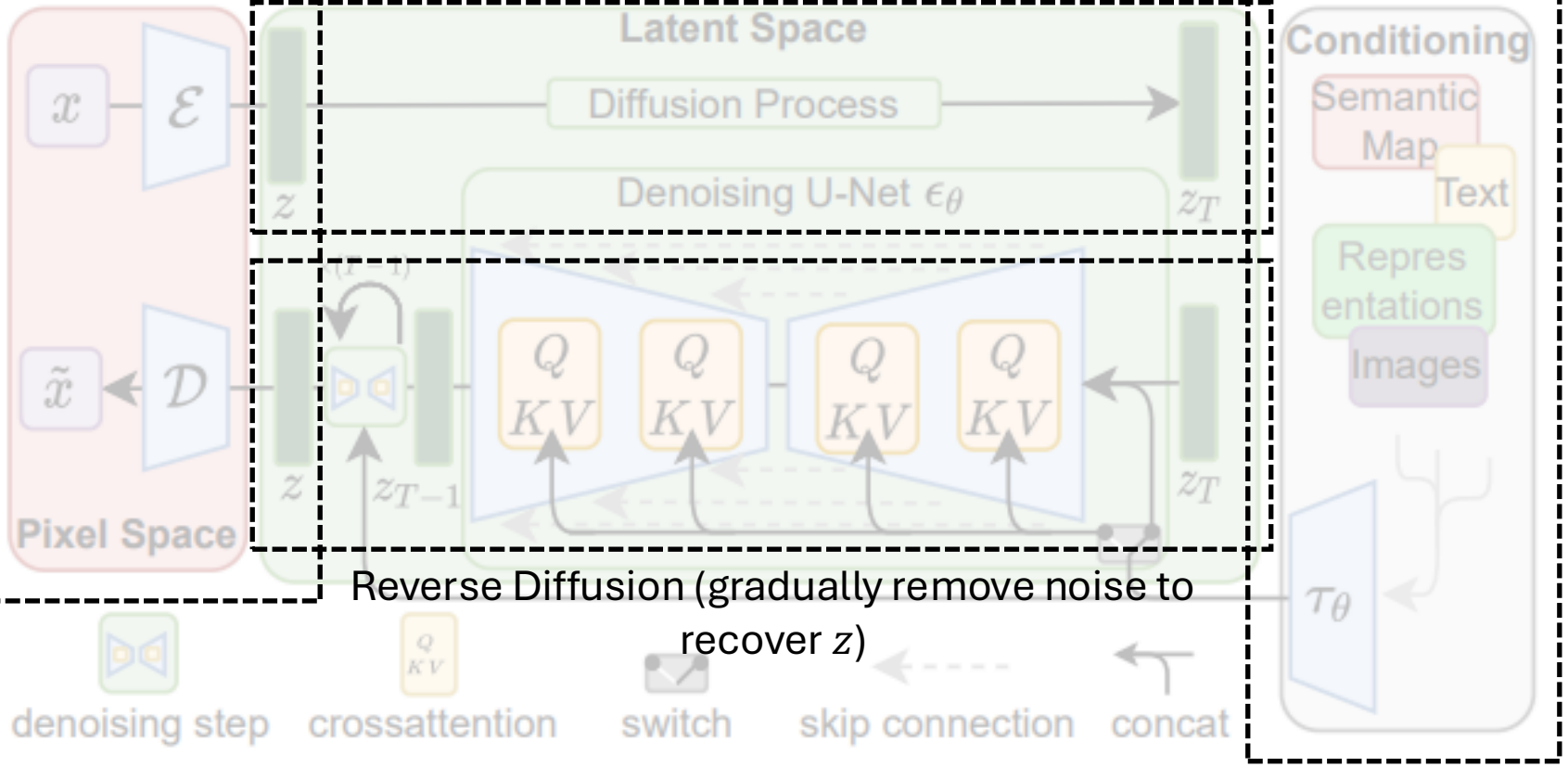
Model that predicts z_t from z_{t+1}

Rombach *et al*, 2022

Latent Diffusion Model

Forward Diffusion (gradually add noise to z)

Autoencoder



Reverse Diffusion (gradually remove noise to recover z)

Represent Conditioning Data

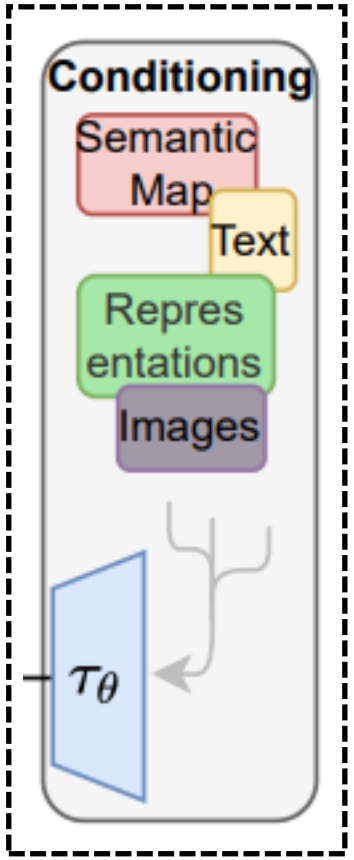
Like DALL-E

Rombach *et al*, 2022

Latent Diffusion Model

Unlike DALL-E:

- $\tau_\theta()$ is an encoder that takes different inputs
- In practice, there are different encoders for each input type
    - For images
    - For text
    - For semantic maps



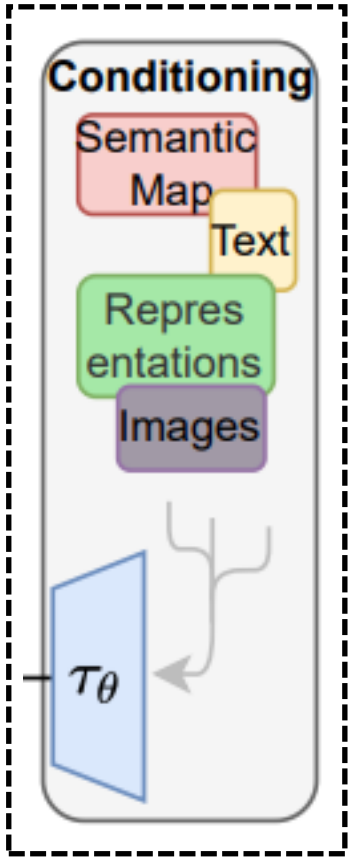
Represent Conditioning Data

Like DALL-E

Rombach *et al*, 2022

Latent Diffusion Model

Conditioning on lower-resolution images (from 64x64 to 256x256)

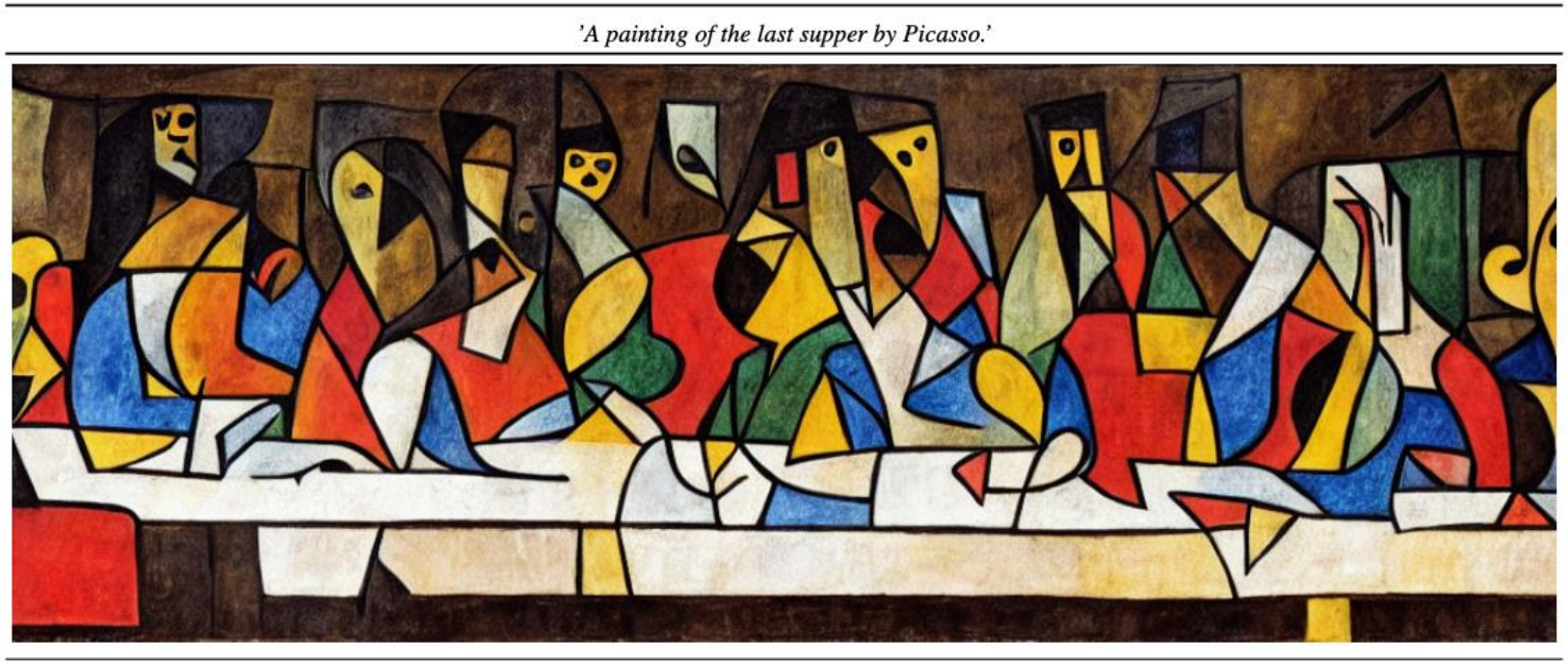


Represent Conditioning Data

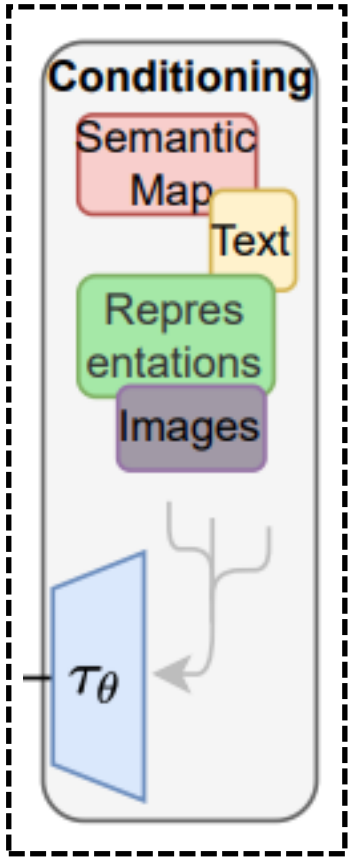
Like DALL-E

Rombach *et al*, 2022

Latent Diffusion Model



Conditioning on text



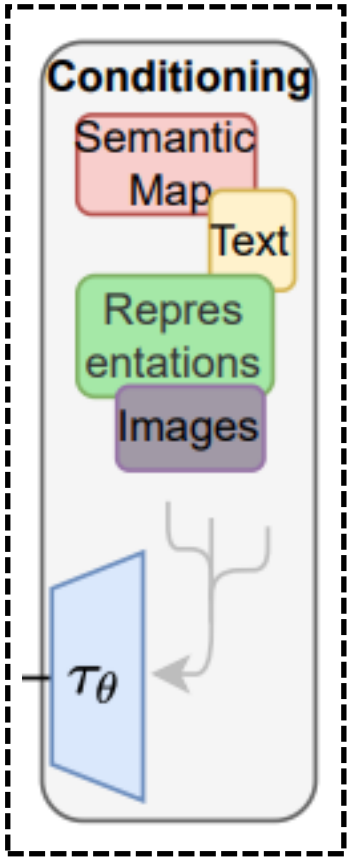
Represent Conditioning Data

Like DALL-E

Rombach *et al*, 2022

Latent Diffusion Model

Conditioning on a semantic map (512x1024)



Represent Conditioning Data

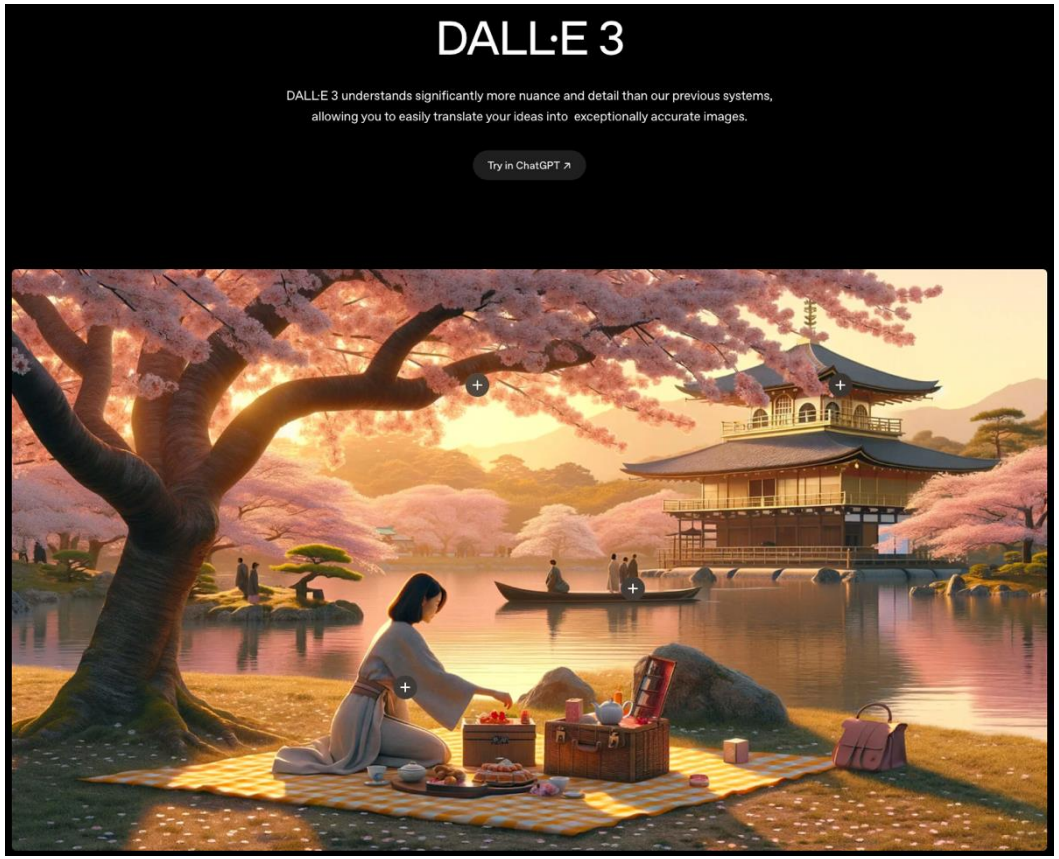
Like DALL-E

Rombach *et al*, 2022

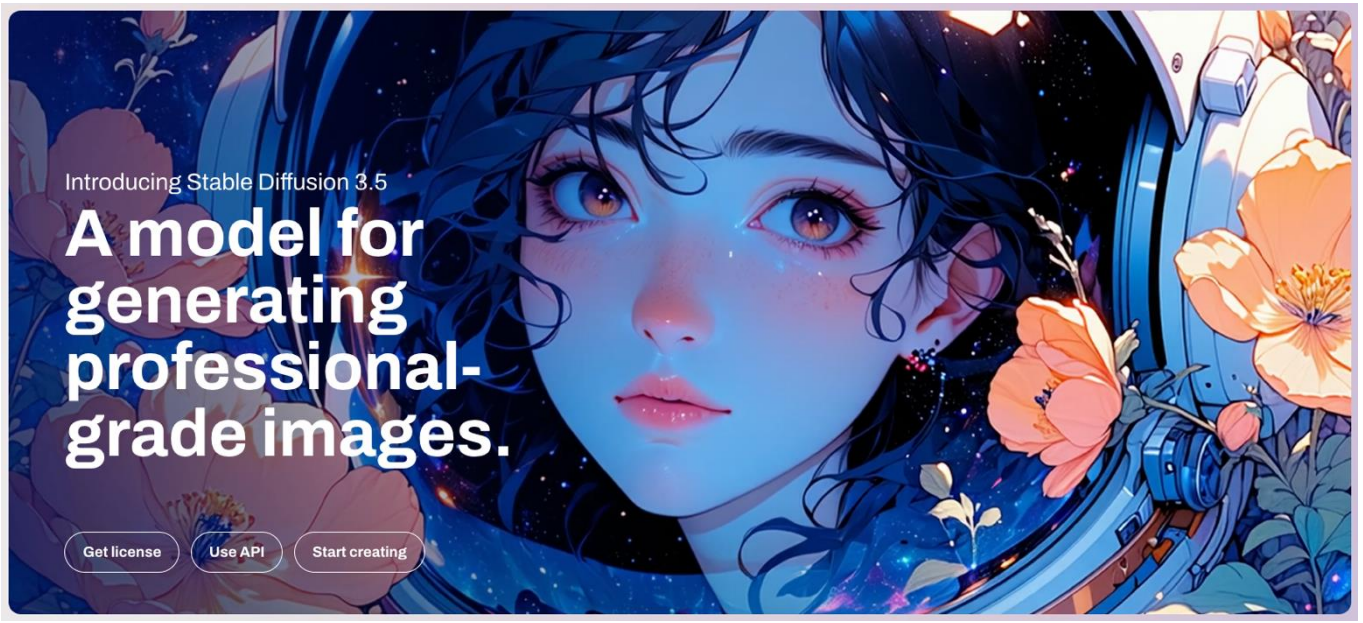
Recap

- Latent diffusion is a combination of three building blocks
  - VAE to reconstruct images from representations
  - Diffusion model (with Attention UNet) to iteratively refine representation
  - Transformer to represent image descriptions
- The model is trained end-to-end using a diffusion objective
- At inference, we encode text/image/map and decode an image

These two models keep improving



DALL-E 3, OpenAI, Sep 2023



Stable Diffusion 3.5, Stability AI, Oct 2024